

Lecture 13: Implicit Differentiation

GOAL: Differentiate functions of the form $F(x, y) = 0$. Use this to find the derivatives of inverse trigonometric functions.

Recall the following:

1. If y is a fun of x , then by the Chain Rule

$$\bullet \frac{d}{dx} [y^2] = \frac{d}{dx} [(y(x))^2] = 2y \frac{dy}{dx}$$

$$\bullet \frac{d}{dx} [\sin(y)] = \frac{d}{dx} [\sin(y(x))] = \cos(y) \cdot \frac{dy}{dx}$$

$$\bullet \frac{d}{dx} [\ln(y)] = \frac{d}{dx} [\ln(y(x))] = \frac{1}{y} \cdot \frac{dy}{dx}$$

2. Taking the derivative does not change equality

$$y = 2x + 1$$

$$y' = 2$$

$\vdots$

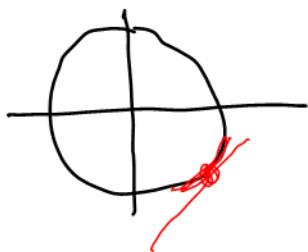
$$y - 2x - 1 = 0$$

$$[y - 2x - 1]' = [0]'$$

$$y' - 2 = 0$$

$$y' = 2$$

Q: How can we find the equation of a line tangent to the unit circle $x^2 + y^2 = 1$?



A: Recognize y is a fun of x

$$x^2 + y^2 = 1$$

$$\frac{d}{dx} [x^2 + y^2] = \frac{d}{dx} [1]$$

$$2x + 2y \frac{dy}{dx} = 0$$

$$2y \frac{dy}{dx} = -2x$$

$$\frac{dy}{dx} = -\frac{2x}{2y}$$

$$\frac{dy}{dx} = -\frac{x}{y}$$

Q: Can we rewrite $\frac{dy}{dx}$ solely as a fun of x ?

A: Sometimes!

$$\begin{aligned} x^2 + y^2 &= 1 \\ y^2 &= 1 - x^2 \\ y &= \pm\sqrt{1-x^2} \end{aligned}$$

$$\frac{dy}{dx} = -\frac{x}{y} = -\frac{x}{\pm\sqrt{1-x^2}} = \begin{cases} -\frac{x}{\sqrt{1-x^2}} & \text{if } y > 0 \\ \frac{x}{\sqrt{1-x^2}} & \text{if } y < 0 \end{cases}$$

$x^2 + y^2 = 1$ is an example of an implicit function and the process of finding $\frac{dy}{dx}$ is called implicit differentiation.

Explicit Functions

$$y = f(x)$$

$$y = 2x + 1$$

$$y = \frac{1}{x}$$

$$y = e^{x^2}$$

$$y = \sin(x)$$

Implicit Functions

$$F(x, y) = 0$$

$$x^2 + y^2 - 1 = 0$$

$$xy - 1 = 0$$

$$e^{xy} + x^2 = 0$$

$$y \sin(xy) - \pi = 0$$

Ex 1

Find the slope of the line tangent to the elliptic curve

$$y^2 = x^3 - x + 1$$

at $(x, y) = (0, 1)$

$$\begin{aligned} y^2 &= x^3 - x + 1 \\ 1^2 &= 0^3 - 0 + 1 \\ 1 &= 1 \checkmark \end{aligned}$$

$$y^2 = x^3 - x + 1$$
$$\frac{d}{dx}[y^2] = \frac{d}{dx}[x^3 - x + 1]$$

$$2y \cdot \frac{dy}{dx} = 3x^2 - 1$$

$$\frac{dy}{dx} = \frac{3x^2 - 1}{2y}$$

$$\left. \frac{dy}{dx} \right|_{(x, y) = (0, 1)} = \frac{3(0)^2 - 1}{2(1)} = -\frac{1}{2}$$

Example 2 Find $\frac{dy}{dx}$ given the algebraic curve

$$\frac{d}{dx}(y^2) \neq y \frac{d}{dx}(x^2) \quad y^3 + yx^2 = x^3 + xy^2$$

$$\rightarrow \frac{d}{dx}[y^3 + yx^2] = \frac{d}{dx}[x^3 + xy^2]$$

$$3y^2 \frac{dy}{dx} + \left[\frac{dy}{dx}(x^2) + y(2x) \right] = 3x^2 + [(1)y^2 + x(2y)\frac{dy}{dx}]$$

$$3y^2 \left[\frac{dy}{dx} \right] + x^2 \frac{dy}{dx} + 2xy = 3x^2 + y^2 + 2xy \frac{dy}{dx}$$



$$3y^2 \frac{dy}{dx} + x^2 \frac{dy}{dx} - 2xy \frac{dy}{dx} = 3x^2 + y^2 - 2xy$$

$$[3y^2 + x^2 - 2xy] \frac{dy}{dx} = 3x^2 + y^2 - 2xy$$

$$\frac{dy}{dx} = \frac{3x^2 + y^2 - 2xy}{3y^2 + x^2 - 2xy}$$

Example 3 Find y' given $xe^y = 9y$

$$[xe^y]' = [9y]'$$

$$(1) e^y + xe^y y' = 9y'$$

$$xe^y y' - 9y' = -e^y$$

$$[xe^y - 9]y' = -e^y$$

$$y' = \frac{-e^y}{xe^y - 9}$$

$$y' = \frac{e^y}{9 - xe^y}$$

Example 4 Find $\frac{dy}{dx}$ given $10 \sin(2x+3y) = 11xy$

$$\frac{d}{dx} [10 \sin(2x+3y)] = \frac{d}{dx} [11xy]$$

$$\underline{10 \cos(2x+3y) \left[2 + 3 \frac{dy}{dx} \right]} = 11y + 11x \frac{dy}{dx}$$

$$20 \cos(2x+3y) + 30 \cos(2x+3y) \frac{dy}{dx} = 11y + 11x \frac{dy}{dx}$$

$$30 \cos(2x+3y) \frac{dy}{dx} - 11x \frac{dy}{dx} = 11y - 20 \cos(2x+3y)$$

$$\left[30 \cos(2x+3y) - 11x \right] \frac{dy}{dx} = 11y - 20 \cos(2x+3y)$$

$$\frac{dy}{dx} = \frac{11y - 20 \cos(2x+3y)}{30 \cos(2x+3y) - 11x}$$

Derivatives of $\sin^{-1}$ and $\tan^{-1}$

Ex Find $\frac{dy}{dx}$ given $y = \sin^{-1}(x)$

Let
 $a = \sin^{-1}x$
 $\sin(a) = \frac{x}{1}$



$$\sin(y) = x$$

$$\cos(y) \cdot \frac{dy}{dx} = 1$$

$$\frac{dy}{dx} = \frac{1}{\cos(y)}$$

$$\frac{dy}{dx} = \frac{1}{\cos(\sin^{-1}(x))}$$

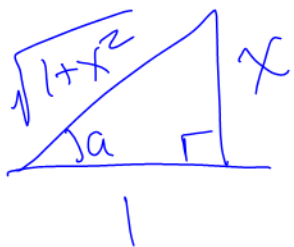
$$\frac{dy}{dx} = \frac{1}{\cos(a)} = \frac{1}{\sqrt{1-x^2}}$$

Ex/ Find $\frac{dy}{dx}$ given $y = \tan^{-1} x$

Let

$$a = \tan^{-1} x$$

$$\tan(a) = \frac{x}{1}$$



$$1^2 + x^2 = z^2$$

$$\tan(y) = x$$

$$\sec^2(y) \frac{dy}{dx} = 1$$

$$\frac{dy}{dx} = \frac{1}{\sec^2(y)} = \frac{1}{[\sec(\tan^{-1}(x))]^2}$$

$$\frac{dy}{dx} = \frac{1}{[\sec(a)]^2} = \frac{1}{1+x^2}$$

Ex/ $\frac{d}{dx} \left(\frac{1}{x} + \frac{1}{y} \right) = \left(\frac{d}{dx} \right)$

$$\frac{d}{dx} \left(\frac{1}{x} \right) = -\frac{1}{x^2}$$

$$-\frac{1}{x^2} - \frac{1}{y^2} \cdot \frac{dy}{dx} = 0$$

$$\left[-\frac{1}{y^2} \right] \frac{dy}{dx} = \frac{1}{x^2}$$

$$\frac{dy}{dx} = -\frac{y^2}{x^2}$$

Ex/ $\sqrt{x} + \sqrt{y} = 4$